INDIVIDUAL MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 6402[1/]

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: 3500 MHz, Mechanically Tunable, 1 Mw Nominal, Pulsed Type, Integral Magnet, Unipotential Cathode.
Outline drawing: Drawing RE38D541
Mounting position: See Note 10
Weight: approximately 65 pounds
Magnet: See Note 9
Input bushing: See Notes 7 and 8

ABSOLUTE RATINGS:

Dependent:	(Note 21)				
Parameters:	ib	Pi	pi	Du	tp
Unit:	a	kW	Mw	---	µs
Maximum:	55	3.7	2.6	0.0016	2.2
Minimum:	40	---	---	---	1.6
					Note 3

Independent:	(Note 22)					
Parameters:	eb	If	tk	VSWR	Tuner torque	Anode T
Unit:	kv	A	sec		inch-ounce	°C
Maximum:	57	50	---	1.5	40	100
Minimum:	---	---	300	---	---	---
		Notes 2 and 15		Note 4		Note 5

Independent:				
Parameters:	Bushing T	Tuner T	Output pressurization	Coolant pressure
Unit:	°C	°C	p.s.i.a.	p.s.i.a.
Maximum:	150	100	55	45
Minimum:	---	---	35	---
	Note 5	Note 5	Note 6	

GENERAL:

Qualification - Required

1/ Formerly 6402/QK437

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.

Method	Requirement or Test	Conditions	Symbol	Limits		Units
				Min	Max	
	Qualification inspection					
1031	High frequency vibration	No voltages	---	---	---	---
---	Shock test	G=15, Note 11	---	---	---	---
4223	Mechanical tuning fatigue	Note 14	---	500	---	cycles
4311	Pushing factor	Note 1	ΔF	---	Note 1	---
4027	Temperature coefficient	Anode temperature 40° to 70°C; F2, Note 5	ΔF/ΔT	---	-0.25	MHz/°C
	Quality conformance inspection, part 1					
---	Tuner driver torque	Notes 12 and 13	Torque	---	30	in-oz
---	Heater voltage	IF=43A	Ef	7.0	9.5	V
---	Standing wave ratio	VSWR=1.1 max.	---	---	---	---
4303	Heater-cathode warmup time	tk=300 seconds (max); Notes 10 and 15	---	---	---	---
4304	Pulse characteristics	tp=1.8 to 2.2 μs; Du=0.0014; Notes 16 and 17	---	---	---	---
4250	Power output	Test B; VSWR=1.1 (max) F1=3430 MHz F2=3500 MHz F3=3570 MHz	PO	980	---	W
---	Average anode current	Ib 70 mAdc	---	---	---	---
4306	Pulse voltage	F1, F2, F3	epy	42	52	kv
4223	Resettability	Anode T, 40° to 80°C, Notes 5 and 16	F	3430	3570	MHz
	Quality conformance inspection, part 2					
4003	Pressurizing	55 psia (min)	---	---	---	---
1031	Low frequency vibration	No voltage	---	---	---	---
4308	RF bandwidth	F2, Note 18	BW	---	3	MHz
4310	Pulling factor	F2	ΔF	---	18	MHz
4315	Stability	F2; VSWR=1.5 Notes 19 and 20	Arcs	---	1	%
	Quality conformance inspection, part 3					
---	Life test	Group D Oscillator (1)	t	250	---	hours
---	Life test end point		PO	784	---	W

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.

NOTES:

1. The pushing factor shall be proposed by the contractor and shall be subject to the approval of the procuring activity.

2. Heater surge current shall not exceed 60 r.m.s. amperes.

3. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall shall be among the more important considerations. The pulse characteristics listed under method 4304 shall be those obtained with modulators used in manufacturing processes and shall not be necessarily applicable with field type modulators. The manufacturer shall be consulted regarding pulse characteristics as related to the specific application.

4. As with any microwave oscillator, frequency skipping or unstable operation may be encountered at some phase position when the mismatch occurs at the end of a "long" line.

5. The temperature shall be measured at the point shown on Drawing RE38D541.

6. The gas used in pressurization shall provide insulating properties at least equal to that of dry air at the pressures indicated.

7. See the installation details shown on Drawing RE38D541.

8. The high voltage bushing shall be immersed in a liquid insulating medium with properties equivalent to Esso Univolt 35 oil.

9. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. Ferromagnetic materials shall not be brought within 6 inches of the tube magnet, except at the ends where the clearance may be 3 inches. Energized magnets shall not be permitted, at any time, closer than 12 inches from the tube magnet.

10. The tube shall be mounted with the cathode vertical during test.

11. The magnetron shall be mounted on a test plate and dropped five times on each of three mutually perpendicular axes parallel to the reference planes shown on Drawing RE38D541. The shock pulse shall have a duration of approximately 11 milliseconds as measured at the quarter amplitude points of the acceleration shock wave.

12. This test shall be conducted at normal operating conditions.

13. The tuner drive mechanism shall not be "set" against either mechanical stop.

14. The tuner mechanism shall be cycled between 20 and 90 turns measured clockwise from the mechanical stop. The turns shall be counted at the point of drive.

15. Specifications for heater operation shall be supplied in operating instructions accompanying each tube.

16. This test shall be conducted only under one set of conditions within the limits stated for the oscillation test specified herein.

17. Typical pulse characteristics for Oscillation (1) shall be as follows:

    trc: Approximately 0.15 μs (measured 20 to 85 percent)
    tfc: Approximately 0.30 μs (measured 0 to 85 percent)
    trv: Approximately 0.40 μs (measured 20 to 85 percent)
    tfv: Approximately 2.00 μs (measured 0 to 85 percent)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.

Voltage or current spike, ripple, inverse and post pulse oscillation shall be at a minimum. The user shall consult the manufacturer if pulse characteristics differing appreciably from those listed above are employed. Tfc means time of fall of current and tfv means time of fall of voltage.

18. Stability shall not be measured under this test. The RF bandwidth shall be within the limits specified herein when a VSWR of 1.5/1.0 is introduced in the load at a distance of approximately 0.5 meters from the magnetron coupling flange, the phase being adjusted for the widest spectrum.

19. Under conditions of oscillation specified herein, the number of arcs shall not exceed the specified percentage of total pulses applied during the last 5 minutes of a test interval and shall not exceed 15 minutes. The arcs shall be counted by an electronic arc detector with biased diode voltage set 10 percent above cutoff.

20. Two observations shall be made, for load phase positions corresponding to maximum power and to minimum power.

21. Caution: The dependent absolute ratings are interrelated, and it shall not necessarily follow that combinations of ratings can be attained simultaneously.

22. The independent absolute ratings shall not be exceeded if the specified life is to be obtained. These ratings are limiting values above which the serviceability of any individual tube may be impaired.

User activities:

Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N824-30)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R00

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of th
ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided
taining information on the use of this specification which will insure that suitable products can be procured
minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.
lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT US

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are a
tional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD, FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:40Z
Check the source to verify that this is the current version before use.